

Customer Churn Analysis & Prediction

A Data-Driven Retention Strategy Report

Prepared for: Telecom Client (Illustrative Engagement)

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Dataset: IBM Telco Customer Churn (7,043 customers)

Executive Summary

This report analyzes customer churn for a telecommunications provider using a dataset of 7,043 customers, 26.5% of whom have churned. The goal is to identify why customers leave, predict which current customers are at risk, and recommend concrete retention actions.

Top Findings

- Contract type is the single strongest driver of churn. Month-to-month customers churn at 42.7%, compared to just 2.8% for two-year contract holders — a more than 15x difference. Most of the benefit comes from moving customers off month-to-month entirely, not from maximizing contract length.
- Three risk factors compound independently: Contract type, Internet Service type (Fiber optic), and Payment Method (Electronic check) each increase churn risk on their own, and their effects stack when combined. Customers with Fiber optic service on a month-to-month contract churn at 54.6% — the highest-risk segment identified.
- Churn is front-loaded. Nearly half of all churn happens within a customer's first 12 months, and this effect is strongest among month-to-month customers specifically.
- Add-on bundling is a strong, actionable retention lever. Customers with zero add-ons churn at over 50%; those with all six churn at just 5%. Online Security and Tech Support are far more protective than Streaming TV/Movies, despite streaming being more popular.
- A predictive model can identify at-risk customers with strong accuracy (ROC-AUC $\approx$ 0.82–0.84) and, importantly, can explain the specific reasons behind each customer's risk score — enabling tailored retention offers rather than one-size-fits-all outreach.

Primary Recommendations

- Use month-to-month as a low-friction entry point, but introduce incentives (discounts, bundled add-ons) to convert customers to a one-year contract after their first few months, once they've experienced the service.
- Prioritize retention outreach for customers who fall into more than one risk category simultaneously (Fiber optic + month-to-month + Electronic check), as this combination represents the highest-risk segment.

- Encourage adoption of Online Security and Tech Support specifically — not streaming services — as these carry the strongest, evidence-backed retention benefit.
- Encourage a switch from Electronic check to an automatic payment method (bank transfer or credit card), which is independently associated with lower churn.
- Deploy the predictive model to flag at-risk customers proactively, using its individual explanations to match each customer with the most relevant retention offer.

1. Business Context & Objective

The dataset represents a telecommunications provider offering phone and internet service, modeled on a realistic subscription business where retaining existing customers is significantly more cost-effective than acquiring new ones. The provider is losing approximately 27% of its customers to churn and needs to understand who is at risk, why, and what actions would meaningfully reduce that risk.

The objective of this analysis is threefold: (1) identify which customer attributes are most strongly associated with churn, (2) build a model capable of scoring existing customers by churn risk, and (3) translate both sets of findings into specific, actionable retention recommendations.

2. Data Overview & Preparation

The dataset contains 7,043 customer records across 21 original features, covering demographics (gender, senior citizen status, partner/dependents), account details (tenure, contract type, billing method), subscribed services (phone, internet type, and six optional add-ons), and billing amounts, alongside the churn outcome.

Data Quality

The dataset required minimal cleaning: no missing values or duplicate records were present. One data type issue was addressed: the TotalCharges field was stored as text rather than a numeric type, and was converted accordingly prior to analysis. Categorical inconsistencies were also addressed: the "No internet service" category, which appeared redundantly across six separate add-on columns (Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, Streaming Movies), was standardized to "No" during feature preparation for modeling, after being deliberately preserved during exploratory analysis to first confirm it represented a single duplicated customer segment rather than six independent signals.

The target variable, Churn, is imbalanced (73.5% No, 26.5% Yes), which informed both the choice of evaluation metrics (precision, recall, and ROC-AUC rather than accuracy alone) and modeling approach (class-weighted models, stratified train/test splitting).

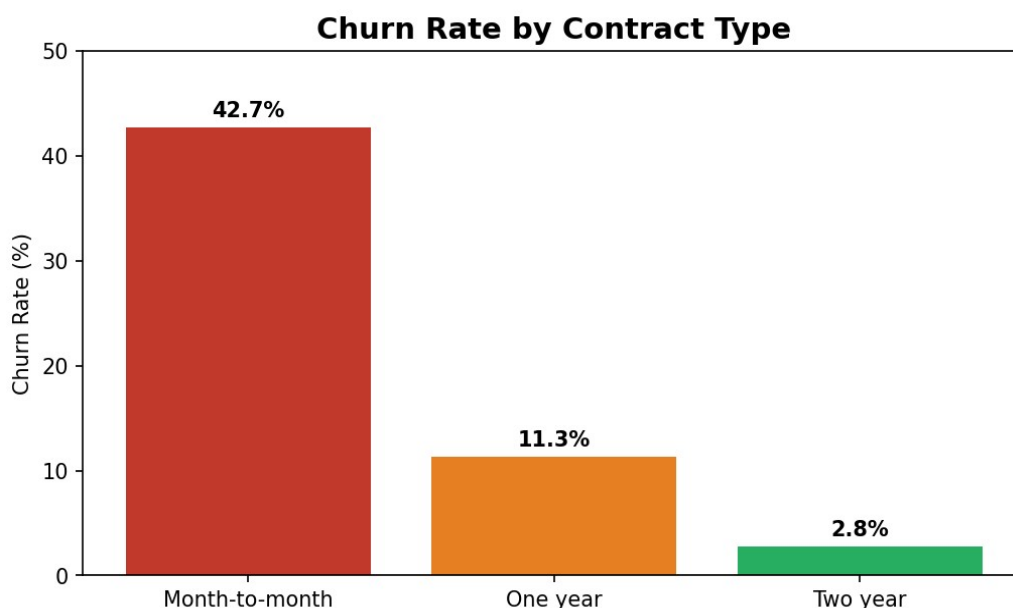
3. Key Findings from Exploratory Analysis

Note: All relationships described below are statistical associations identified in historical data. They indicate strong, consistent patterns but do not by themselves establish direct causation; contributing factors such as service quality, pricing perception, or competitive pressure cannot be fully isolated from this dataset alone.

3.1 Contract Type is the Strongest Single Driver of Churn

Contract Type	Churn Rate
Month-to-month	42.7%
One year	11.3%
Two year	2.8%

Month-to-month customers churn at more than 15 times the rate of two-year contract holders. The steepest drop occurs between month-to-month and one-year contracts (a 31.4-point decrease); the further drop to two-year contracts is comparatively small (8.5 points). This indicates the primary retention benefit comes from moving customers into any fixed-term commitment, not specifically the longest one available. Month-to-month customers also represent the largest segment overall (54% of the base), making this the single largest source of both revenue and churn exposure.



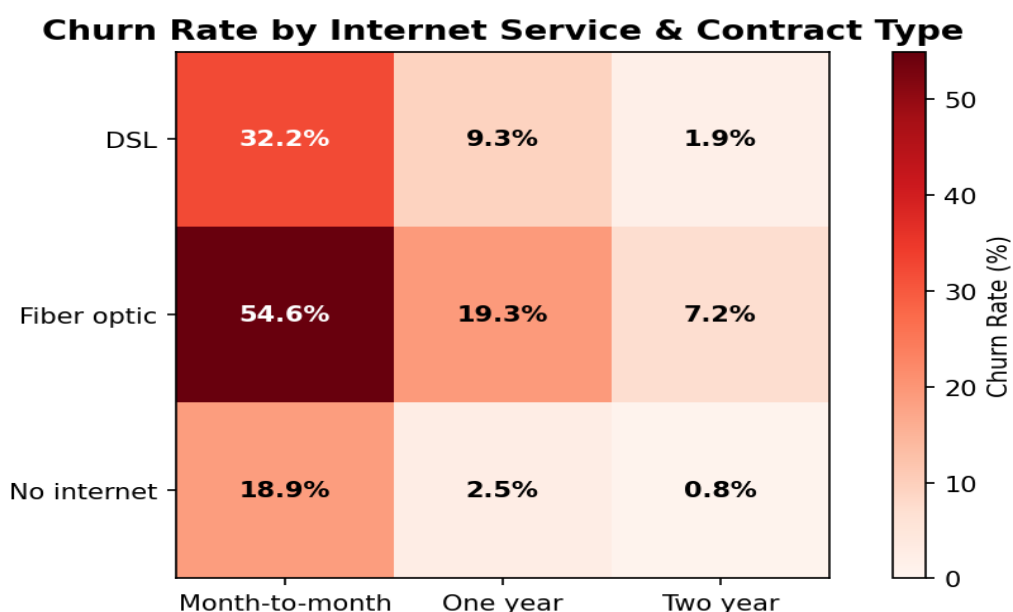
3.2 Three Risk Factors Compound Independently

Contract type, Internet Service type, and Payment Method were each tested against one another to determine whether their individual effects on churn were genuinely independent or simply reflected overlapping customer populations. All three were confirmed independent and additive:

Risk Factor Combination	Churn Rate
Fiber optic + Month-to-month	54.6%
Electronic check + Month-to-month	53.7%
Fiber optic + Electronic check	53.2%
Two-year + Any payment method	0.79%–7.74%

Each risk factor remains elevated within every level of the other two — for example, Fiber optic customers churn more than DSL customers at every contract length, not just in aggregate. This confirms these are three genuinely independent, compounding risk factors. Customers exhibiting more than one simultaneously represent the highest-priority segment for retention outreach, likely accounting for a disproportionate share of total churned revenue.

Notably, Electronic check's elevated churn is specific to that payment method rather than a general "manual payment" effect: Mailed check (also manual) tracks closely with the two automatic payment methods across every contract and internet service type, while Electronic check remains a consistent outlier.



3.3 Churn Risk is Front-Loaded, Especially for Month-to-Month Customers

Tenure (months)	Churn Rate
0–12	47.4%
13–24	28.7%
25–48	20.4%
49–72	9.5%

Customers who churn have a median tenure of roughly 10 months, versus 38 months for retained customers. However, this protective effect of tenure is not uniform: it is strong and steadily declining among month-to-month customers (51.4% churn in month 0–12, down to 26.0% by month 49–72), but flat among one-year contract customers and negligible among two-year customers. This means tenure milestones are most useful as a retention trigger specifically for customers who have not yet committed to a fixed-term contract.

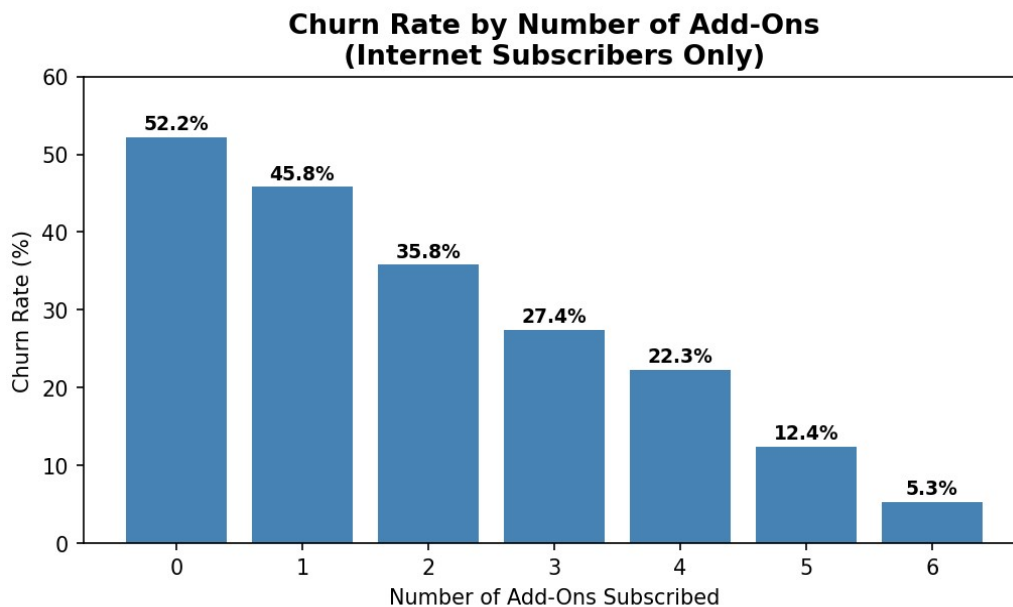
3.4 Add-On Services Meaningfully Reduce Churn — But Not Equally

Add-On Service	Churn Reduction (Yes vs. No)
Online Security	27.2 points
Tech Support	26.5 points
Online Backup	18.4 points
Device Protection	16.6 points
Streaming Movies	3.7 points
Streaming TV	3.5 points

All six add-ons are associated with reduced churn, but the size of the effect varies substantially. Online Security and Tech Support nearly halve churn risk; Streaming TV and Streaming Movies — despite being the most widely adopted add-ons — have only a marginal protective effect. Considered together, customers with zero add-ons churn at over 50%, compared to roughly 5% for

customers with all six, confirming that overall bundling is a strong retention lever, with security and support services carrying the most weight.

This has a direct implication for incentive design: a retention strategy built around free trials of Online Security or Tech Support is likely to be far more effective than one centered on streaming services, despite streaming's higher current adoption.



3.5 Household Structure is a Meaningful Secondary Signal

Household Type	Churn Rate
Neither partner nor dependents	34.2%
Partner only	25.4%
Dependents only	21.3%
Partner and dependents	14.2%

Partner and dependent status are weak predictors of churn individually, but combined, they reveal a clear pattern: churn declines steadily as household commitment increases, likely reflecting higher switching costs (shared billing, more people affected by a service change) for more established households. While this cannot be directly influenced by the business, it is a useful targeting signal for identifying higher-risk customers.

4. Predictive Modeling

Two models were built to predict individual customer churn risk: a logistic regression baseline and a Random Forest classifier. Both were trained on an 80/20 stratified train-test split (preserving the 73.5%/26.5% churn ratio in both sets) and adjusted for class imbalance using balanced class weighting.

4.1 Model Comparison

Metric	Logistic Regression	Random Forest (default threshold)	Random Forest (tuned threshold)
Accuracy	0.74	0.79	—
Recall (Churn)	0.79	0.50	0.78
Precision (Churn)	0.51	0.64	0.50
ROC-AUC	0.84	0.83	0.83

Both models show a similar underlying ability to distinguish churners from non-churners (ROC-AUC $\approx$ 0.83–0.84). At default settings, Random Forest appeared more conservative (higher precision, lower recall), while logistic regression favored catching more churners at the cost of more false alarms. Since failing to identify a customer who will churn is generally costlier to the business than a false alarm, the Random Forest's classification threshold was lowered from 0.5 to 0.25, at which point its performance closely matched logistic regression. 5-fold cross-validation confirmed the Random Forest's performance is stable across different data splits (ROC-AUC range: 0.805–0.839, mean 0.822).

Given the near-identical performance of both models once tuned to the same business priority, either is a defensible choice for deployment. Logistic regression offers simpler, built-in interpretability; Random Forest can capture more complex, conditional relationships between features (such as tenure's effect varying by contract type) and, paired with additional interpretability tooling, offers equally clear per-customer explanations. This report proceeds with the Random Forest model as the primary recommendation, given its slightly better handling of the compounding risk factors identified in the exploratory analysis.

4.2 What Drives the Model's Predictions

The model's most influential features closely mirror the exploratory findings: tenure, internet service type, contract length, and payment method are the strongest predictors of individual churn risk, in both models. Two secondary findings are worth noting for transparency:

- Streaming add-ons (TV and Movies) showed an inconsistent effect direction between the exploratory analysis and the logistic regression model, reflecting their genuinely weak, near-zero relationship with churn overall rather than a meaningful reversal.
- Household type showed a more nuanced pattern than the bivariate analysis alone suggested. Customers with neither a partner nor dependents, and customers with a partner only, both retained a meaningful, independent increase in churn risk even after accounting for other features. However, the added protective benefit of having both a partner and dependents together (compared to having dependents alone) was not confirmed once contract type and tenure were included directly — suggesting that specific combination's bivariate advantage is substantially explained by those stronger, more directly measurable factors, rather than household structure itself.

5. Understanding Individual Customer Risk

Beyond identifying which customers are likely to churn, the model can explain why a specific customer is at risk — breaking their prediction down into the individual factors driving it up or down. This moves beyond a single risk score to a clear, actionable explanation for each customer, and is consistent with the compounding-risk-factor pattern identified in the exploratory analysis.

Illustrative Examples

Customer A (predicted risk: near 0%): A long-tenured customer on a two-year contract with Online Security and Tech Support subscriptions. These factors combine to make this an extremely low-risk customer, even though their Fiber optic service works mildly against them.

Customer B (predicted risk: 79%): A senior citizen with no long-term contract, paying by Electronic check, on Fiber optic service. Each factor independently increases risk, and together they compound into a high-risk profile.

These examples illustrate a consistent theme across this analysis: churn risk is rarely driven by a single factor. It typically results from multiple risk factors — contract type, internet service, payment method — stacking together.

Business Value

This level of detail allows retention efforts to be tailored to the specific reason a customer is at risk, rather than applying the same generic offer to everyone flagged as high-risk. A customer whose primary risk driver is lack of a long-term contract may respond best to a contract-conversion incentive; a customer whose risk stems mainly from payment method may respond better to an autopay incentive instead.

6. Business Recommendations

- Convert month-to-month customers gradually: use the plan as a low-friction acquisition channel, then introduce a contract-conversion incentive (discount or bundled add-on) after the first few months of service, once the customer has experienced the product firsthand.
- Prioritize customers with compounding risk factors: those on Fiber optic, month-to-month contracts, and/or Electronic check payment simultaneously represent the highest-priority segment, independent of which single factor is examined.
- Promote Online Security and Tech Support specifically, not streaming services, as the primary add-on incentive — this recommendation is directly evidence-based, given the far larger measured effect of these two services on retention.
- Encourage a shift away from Electronic check toward automatic payment methods (bank transfer or credit card), which are independently associated with meaningfully lower churn.
- Target the first 12 months of the customer lifecycle for retention attention, particularly for month-to-month customers, where churn risk is highest and most responsive to early intervention.
- Deploy the predictive model operationally (e.g., via a live dashboard) to score current customers by risk and surface the specific factors behind each score, enabling retention teams to act with targeted, rather than generic, outreach.

7. Limitations & Next Steps

- This analysis is based on a single historical snapshot of customer data rather than ongoing monitoring; churn drivers may shift over time as pricing, competition, or service quality change.
- The dataset does not include direct measures of service quality, customer support interactions, or competitor pricing, which likely contribute to some of the patterns observed (e.g., Fiber optic's elevated churn) but cannot be directly tested here.
- Relationships identified throughout this report are statistical associations, not confirmed causal mechanisms; recommended interventions should ideally be validated through small-scale testing (e.g., a pilot contract-conversion incentive) before full-scale rollout.
- Future work could include: testing additional model types (e.g., gradient boosting) for further performance gains, deploying a live-updating dashboard for ongoing risk monitoring, and incorporating customer feedback or support-ticket data to test the underlying causes behind key findings such as Fiber optic's elevated churn.